Power Distribution Overview

Our system uses a **12V power supply** as the main source. Power is distributed in two ways:

- **Directly** to components that operate at 12V (like the robotic arm motors)
- **Through buck converters** to step down voltage for lower-voltage components

Why Buck Converters Are Needed

A **buck converter** reduces voltage safely and efficiently.
We use them for:

- **Platform motors (6V):** Convert 12V → 6V to prevent damage
- **Raspberry Pi (5V):** Convert 12V → 5V to avoid failure

Without buck converters, components would overheat or be permanently damaged.

Current Calculation

Total current required:

- Platform motors: 1 A
- Raspberry Pi: 3 A
- Robotic arm motors: 16.2 A
- Camera: 1 A
- LCD: 1 A
- Arduino: 1.5 A
- Motor driver: 0.1 A
- Buck converters: 4.2 A

Total = 25.3 A

Power Supply Selection

To ensure safe operation and handle peak loads, we choose:
12V power supply rated ~30A or higher

Conclusion

- 12V supply powers the system
- Buck converters ensure correct voltage levels
- Proper current calculation ensures reliability and safety

12 V
12 V
6 V
6 V